REPORT 2

Exploratory Data Analysis Report

Sales patterns, store performance, promotions and customer behaviour

Project	Rossmann Store Sales
Stage	Exploratory Data Analysis
Period covered	2013-01-01 to 2015-07-31
Report date	12 November 2026

5.87 bn	5,773.82	+38.77%	0.824
Total sales	Average daily sales	Sales uplift with promotion	Customer-sales correlation

1. Dataset Overview

Metric	Value
Total records	1,017,209
Total stores	1,115
Total columns	25
Date range	2013-01-01 to 2015-07-31

2. Sales Summary

Metric	Value
Total sales	5,873,180,623.00
Average sales	5,773.82
Median sales	5,744.00
Maximum sales	41,551.00

3. Customer Summary

Metric	Value
Average customers	633.15
Maximum customers	7,388

4. Store Performance

Metric	Value
Total stores	1,115
Best performing store type	b
Highest sales month	3 (March)

5. Promotion Analysis

Metric	Value
Average sales without promotion	5,929.41
Average sales with promotion	8,228.28
Percentage difference	38.77%

6. Correlation Analysis

Variable Pair	Correlation
Customers vs Sales	0.824

7. Business Insights

- Promotion performance was analysed.
- Store type performance was compared.
- Monthly sales patterns were identified.
- Customer and sales relationships were examined.

8. Conclusion

The exploratory analysis identified sales patterns, store performance differences, promotional trends and time-based variations. These findings can support business decisions related to inventory planning, promotions and store performance improvement.